2026

North-West Plastics Manufacturing (Klerksdorp)

MILESTONE 1

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Contents

Client Requirements 2
IP Addressing Plan 3
Topologies 3
Physical Topology: 3
Logical Topology 4
Initial GitHub Repository Link: 7

Client Requirements

1. **Introduction:**

   The project is about developing a functional network that uses a WPA2-PSK authentication, for the North-West Plastics Manufacturing organisation in Klerksdorp. The completed network must permit successful data exchange between appropriate nodes and demonstrate the Wireless Security(WPA2-PSK).

2. **Functional Requirements**

   The client expects a functional network where :

   - All required devices are connected.
   - Routers, switches and end devices are configured.
   - The network passes connectivity tests.

   Cisco Packet Tracer must be used for implementation.

3. **Security Requirements and Constraints**

   - The wireless network must use WPA2-PSK
   - A secure pre-shared password or key must be configured.
   - The IoT sensors must be placed on an isolated network segment.
   - Legacy devices without modern security features must still be accommodated safely.
   - The WPA2-PSK configuration must be tested.
   - Any security related troubleshooting must be documented.

4. **Change request Requirements**

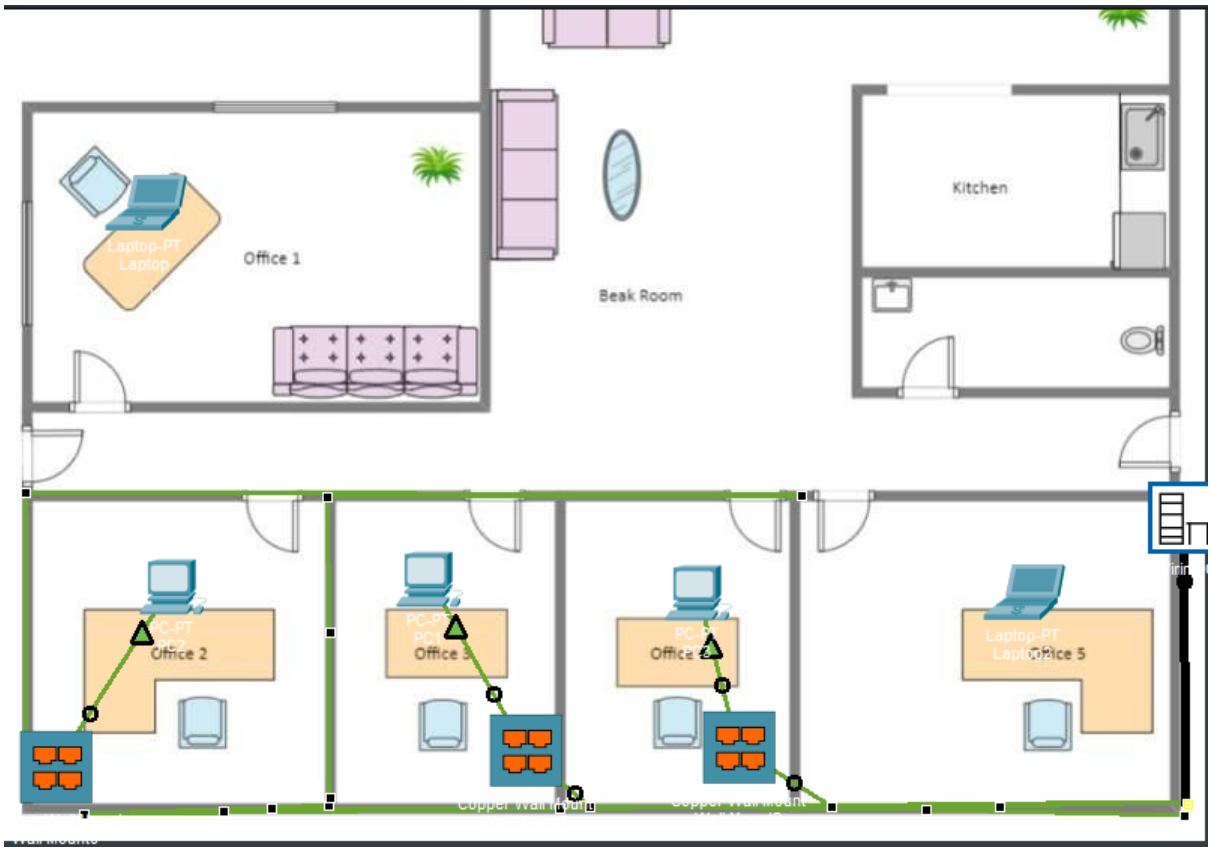
   - The network must include IoT sensors.
   - A separate network segment must be created.
   - Network configuration must be included so that IoT devices can communicate as required.

5. **Performance and reliability requirements**

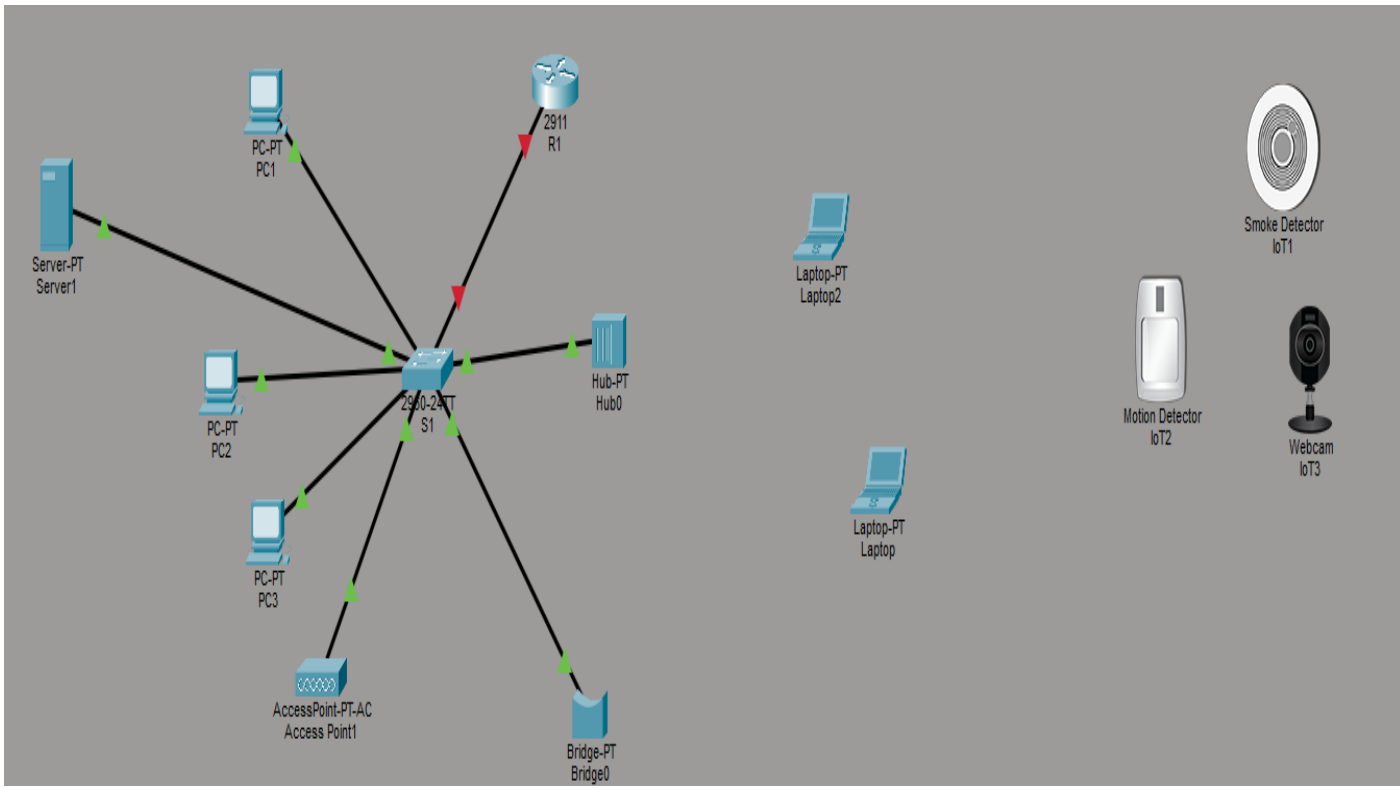
   - Relevant devices must be tested for connectivity
   - Networking configuration must be verified if it works
   - Important troubleshooting recorded must be documented
   - The assigned networking challenge must be verified
   - The Packet Tracer file must open and work correctly

Topologies

Physical Topology:



Logical Topology



IP Addressing Plan

IP Addressing Block: 192.168.23.0/24

Subnet Mask :255.255.255.0

The /24 addressing block is divided into smaller sub nets used to separate the different network segments since the client change request required that the IoT sensors must be added to the operations area and need an isolated segment.

VLAN	Network Purpose	Network Address	Subnet Mask	Default Gateway	Usable Host Range
VLAN 10	Main LAN	192.168.23.0/26	255.255.255.192	192.168.23.1	192.168.23.2-192.168.23.62
VLAN 20	Legacy Devices	192.168.23.64/27	255.255.255.224	192.168.23.65	192.168.23.66-192.168.23.94
VLAN 30	IoT sensors(isolated)	192.168.23.96/27	255.255.255.224	192.168.23.97	192.168.23.98-192.168.23.126
VLAN 40	Network Management	192.168.23.128/28	255.255.255.240	192.168.23.129	192.168.23.130-192.168.23.142

Device	VLAN	IP Adress	Subnet Mask	Gateway
RouterG0/0.10	10	192.168.23.1	255.255.255.192	
RouterG0/0.20	20	192.168.23.65	255.255.255.224	
RouterG0/0.30	30	192.168.23.97	255.255.255.224	
RouterG0/0.40	40	192.168.23.129	255.255.255.240	
PC1	10	192.168.23.10	255.255.255.192	192.168.23.1
PC2	10	192.168.23.11	255.255.255.192	192.168.23.1
Laptop1	10	192.168.23.12	255.255.255.192	192.168.23.1
Laptop2	10	192.168.23.13	255.255.255.192	192.168.23.1
API	10	192.168.23.20	255.255.255.192	192.168.23.1
Legacy Device1	20	192.168.23.70	255.255.255.224	192.168.23.65
Legacy Device2	20	192.168.23.71	255.255.255.224	192.168.23.65
IoT sensor1	30	192.168.23.100	255.255.255.224	192.168.23.97
IoT sensor2	30	192.168.23.101	255.255.255.224	192.168.23.97
IoT sensor 3	39	192.168.23.102	255.255.255.224	192.168.23.97
Switch Management	40	192.168.23.130	255.255.255.240	192.168.23.129
Server	10	192.168.23.30	255.255.255.192	192.168.23.1

Initial GitHub Repository Link:

GitHub repository link : https://github.com/Almagry/CMPG325-2026-030-Wireless-Security-